

## CALCUTTA MATHEMATICAL SOCIETY INTERNSHIP PROGRAMME



# Neyman-Pearson Lemma and Its Applications in Machine Learning

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Submitted by

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## Why the lemma matters

In many statistical and machine learning problems, the two types of error do not have the same cost. The Neyman-Pearson framework is designed for exactly this setting.

- ▶ A false alarm may be tolerable, but missing a critical case may be serious.
- ▶ The goal is to control one error first and then optimize the other.
- ▶ This idea is central in high-stakes classification problems.

# Type-I and Type-II Errors

## Type-I error

$$\alpha = \mathbb{P}(\text{Reject } H_0 \mid H_0 \text{ is true})$$

- ▶ False positive
- ▶ Size of the test
- ▶ Kept under a fixed bound

## Type-II error

$$\beta = \mathbb{P}(\text{Fail to reject } H_0 \mid H_1 \text{ is true})$$

- ▶ False negative
- ▶ Missed detection
- ▶ Power equals  $1 - \beta$

| True State | Decision             | Outcome                   |
|------------|----------------------|---------------------------|
| $H_0$ true | Reject $H_0$         | Type-I error ( $\alpha$ ) |
| $H_0$ true | Fail to reject $H_0$ | Correct decision          |
| $H_1$ true | Reject $H_0$         | Correct decision          |
| $H_1$ true | Fail to reject $H_0$ | Type-II error ( $\beta$ ) |

# Formal Statement of the Neyman-Pearson Lemma

## Theorem (Neyman-Pearson Lemma)

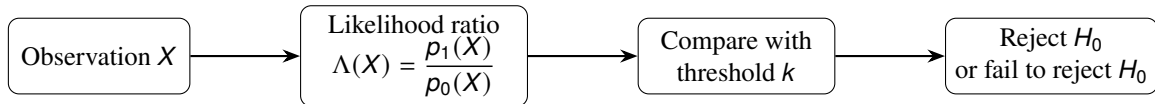
*Let  $P_0$  and  $P_1$  be two simple hypotheses with densities  $p_0$  and  $p_1$  with respect to a common measure. Among all tests of size  $\alpha$ , the most powerful test is given by the likelihood-ratio rule*

$$\Lambda(x) = \frac{p_1(x)}{p_0(x)} \geq k,$$

*where the threshold  $k$  is chosen so that the test has size  $\alpha$ .*

- ▶ Reject  $H_0$  when the evidence against it is large enough.
- ▶ If needed, randomization can be used on the boundary set.

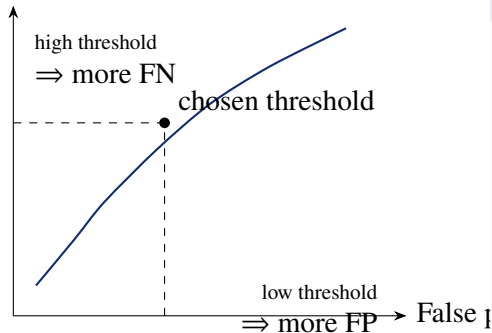
# Decision Process



- ▶ The theorem becomes a threshold-based decision rule.
- ▶ This is the key bridge to machine learning classification.

# ROC / Threshold Trade-off

True positive rate



## Interpretation

- ▶ Lower threshold: higher power, but more false positives.
- ▶ Higher threshold: fewer false positives, but more false negatives.
- ▶ Type-I error corresponds to the false positive rate.
- ▶ Type-II error is linked to  $1 - \text{TPR}$  and missed detections.
- ▶ NP classification chooses an operating point under a constraint.

# Machine Learning Connection

- ▶ Binary classification can be viewed as hypothesis testing.
- ▶ A learner outputs a score or ranking.
- ▶ The decision threshold is chosen to control the protected error.
- ▶ It is closely related to cost-sensitive learning and imbalanced datasets.
- ▶ Precision-recall trade-offs become more important than accuracy alone.

## Why it is useful

- ▶ Decision thresholds
- ▶ Asymmetric costs
- ▶ Precision-recall trade-offs
- ▶ Controlled false positives
- ▶ Practical risk management

# Numerical Example

Consider  $X \sim \text{Bin}(3, p)$  with

$$H_0 : p = 0.2 \quad \text{and} \quad H_1 : p = 0.8.$$

The likelihood ratios are

$$\Lambda(0) = \frac{0.2^3}{0.8^3} = \frac{1}{64}, \quad \Lambda(1) = \frac{0.8 \cdot 0.2^2}{0.2 \cdot 0.8^2} = \frac{1}{4}, \quad \Lambda(2) = 4, \quad \Lambda(3) = 64.$$

Hence a likelihood-ratio test rejects  $H_0$  when  $X \geq 2$ .

$$\alpha = \mathbb{P}_0(X \geq 2) = \binom{3}{2}(0.2)^2(0.8) + (0.2)^3 = 0.104,$$

$$1 - \beta = \mathbb{P}_1(X \geq 2) = \binom{3}{2}(0.8)^2(0.2) + (0.8)^3 = 0.896.$$



# Concrete Applications

- ▶ **Medical diagnosis:** distinguish healthy vs diseased cases with controlled false alarms.
- ▶ **Credit card fraud detection:** reduce missed fraud while limiting rejected genuine transactions.
- ▶ **Network intrusion detection:** flag suspicious traffic without over-alerting on normal activity.
- ▶ **Anomaly detection:** detect rare events while keeping alert volume manageable.

# Research Relevance and Future Scope

- ▶ Modern machine learning uses the NP idea in constrained optimization and fairness-aware decision making.
- ▶ It is relevant when one wants guarantees on a specific error rate rather than only accuracy.
- ▶ Ongoing research includes high-dimensional learning, distribution-free guarantees, and label-noise robustness.
- ▶ It is increasingly relevant for AI safety, medical AI, and risk-sensitive deployment.

# Key Takeaways

- ▶ The Neyman-Pearson lemma gives the most powerful test at a fixed size  $\alpha$ .
- ▶ The likelihood-ratio rule is the core mathematical principle.
- ▶ In machine learning, the same idea becomes threshold-based classification with controlled risk.
- ▶ The framework is useful wherever false positives and false negatives have different costs.

**Thank You**